CS3 L6

Deep Learning

Machine Learning

$$f(X) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

↳ we want to find a function $\hat{f}$ s.t. $Y \approx \hat{f}(X)$ for any observation (X, Y)

x_i = features / variables ; y_i = label = target / outcome variable

eg: linear regression $f(x) = wx + b$

multiple linear regression $f(x_1, \dots, x_n) = \sum(w_i \cdot x_i) + b$

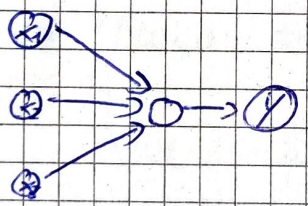
solved with SVM → more complex function

→ Neural Networks:

(a tool to find an approximate function that describes the dataset) / (a glorified line fitting problem)

Training data
Test data
→ Parameters

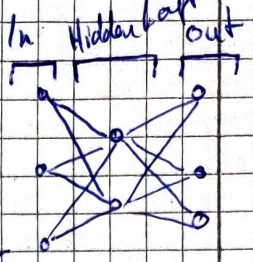
finding the minimum of cost function



$$y_i = f\left(\sum_{j=1}^{N} w_{ij}^{1} x_j^{1}(x) + b_i^{1}\right)$$

Neural Networks

Structure: multilayer perceptron



Neuron: → holds a number $a \in [0,1]$
↳ activation

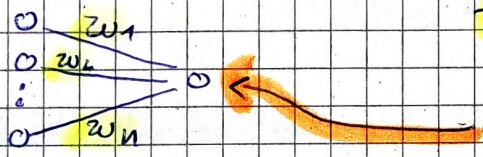
→ Last layer: how much the system thinks the capital is ...

function $f(a_0, \dots, a_n) = \begin{bmatrix} y_0 \\ \vdots \\ y_n \end{bmatrix}$

→ one layer activates the next layer until the last

→ In theory: Input is broken down to minimal subproblem then each next layer represents a combined subproblem, till the output can be predicted

→ Assign weights to the connections of the neuron



$w_1 a_1 + w_2 a_2 + \dots + w_n a_n - X$ → near one if $> X$

X: bias

$\sigma(x) = \frac{1}{1+e^{-x}}$ → near one if > 0

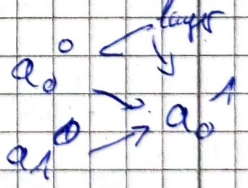
→ Learn: find the right weights and biases

sigmoid σ $[0,1] \ni$

$$\sigma\left(\begin{pmatrix} w_{0,0} & w_{0,1} & \dots & w_{0,n} \\ \vdots & & & \vdots \\ w_{k,0} & w_{k,1} & \dots & w_{k,n} \end{pmatrix} \begin{pmatrix} a_0^0 \\ \vdots \\ a_n^0 \end{pmatrix} + \begin{pmatrix} b_0 \\ \vdots \\ b_n \end{pmatrix}\right)$$

$\vec{a}^1 = \sigma(W \vec{a}^0 + b)$ for

(activation layer)



CSB

Neural Networks

→ Initialize all weights and biases randomly!

→ Cost Function:

$$\text{Cost}(\text{input}) = (\text{out}_1 - \text{real})^2 + \dots + (\text{out}_n - \text{real})^2$$

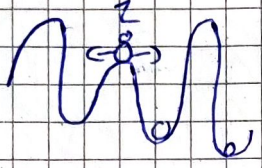
→ Take Ø costs of all training data

In: weigh/biases

Out: costs

Parameters: Training Data

$C(w)$, find input with the minimal costs ($\frac{dC}{dw}(w) = 0$)



Local Minimum: Okay

Global Minimum: Hard!

Which direction decreases $C(x,y)$ most quickly?

↳ Gradient: $\nabla C(x,y)$

(Direction with highest increase)

Algo: Compute $\nabla C(x,y)$

Small step in $-\nabla C$ direction (small = learning rate)

Repeat

⇒ More hidden layers: (+) fit more complex functions

(−) more prone to overfitting

→ Instead of using the sigmoid function, other activation functions can be used eg: Binary Step, Softplus

Softmax, when your output to be the probability of being in particular class (needs at least two classes)

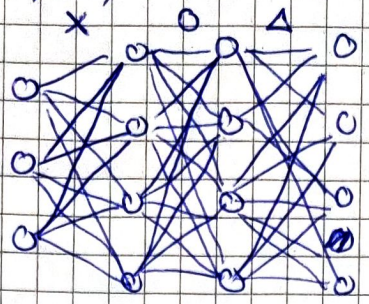
CSB

Neural Networks

biases & # weights

fully connected

3 inputs, 2 hidden layers (4 nodes), 4 outputs



$\times : 3\binom{4}{}() + b \in \mathbb{R}^4$

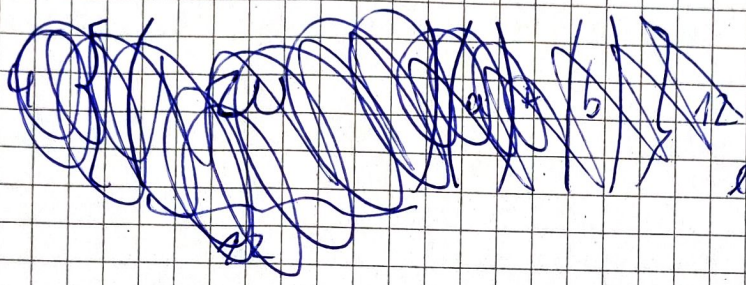
$\circ : 4\binom{4}{}()4 + 1)4$

$\triangle : 4\binom{4}{}()4 + ()4$

weights: $3 \cdot 4 + 4 \cdot 4 + 4 \cdot 4 = 12 + 16 + 16 = 48$

$|\text{Nodes}_{i-1}| \cdot |\text{Nodes}_i| + \ldots$

biases: Σ hidden nodes + Output nodes $= 12$



$w_{k,i} \in \mathbb{R} \quad i \in \{0, \ldots, n\}$

$z^{(l)} = w^{(l)} a^{(l-1)} + b^{(l)}$

$a^{(l)} = g^{(l)}(z^{(l)})$

l is like in $wa + b = z$

$$\begin{pmatrix} w_{0,0} & \cdots & w_{0,n} \\ \vdots & & \\ w_{k,0} & \cdots & w_{k,n} \end{pmatrix} \begin{pmatrix} a_0^0 \\ a_n^0 \end{pmatrix} + \begin{pmatrix} b_0 \\ b_n \end{pmatrix} = z \in \mathbb{R}^n$$

$$w\vec{a} + \vec{b} = \vec{z} \quad \text{or} \quad \forall i \in [1, k] : z^{(i)} = w^{(i)} a^{(i-1)} + b^{(i)}$$

$\hookrightarrow a^l = g^l(z^l)$

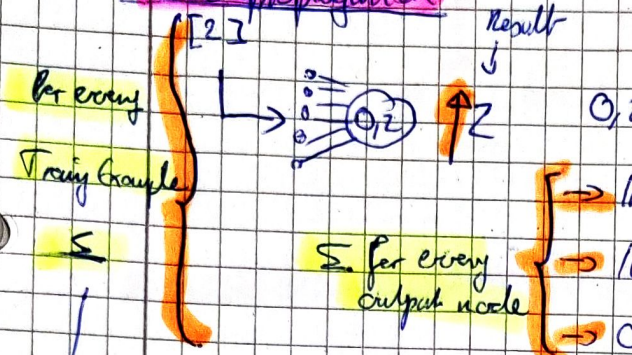
CSS

Cost Function Different Cost function

eg: MSE, L2, L1, Euclidean Distance

Goal reduce distance between predicted and real!

Backpropagation



→ Do this in Mini Batches (Training Data)

$$\frac{\partial C}{\partial w^L} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{\partial C_k}{\partial w^L}$$

↳ small batches => unstable, big batches = won't fit in GPU

→ Learning Rate

Higher learning rate: unstable training and poor solutions

& lower learning rate: results in slower training and sub-optimal solutions

→ There are faster Algorithms

→ Regulation Tips

Problems: Overfitting, fails to learn, collapse to 0, gradient explosion

Tip: • start with normalized data

• Dropout Layer: while training randomly drops connections with certain probs

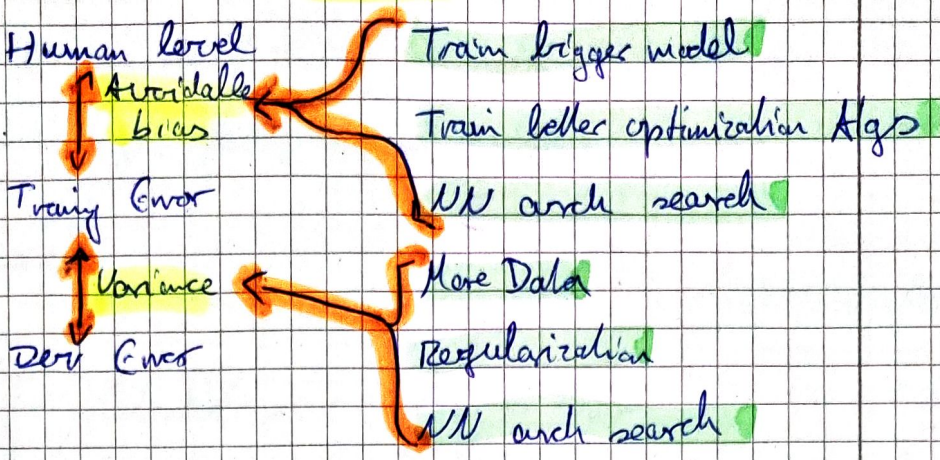
• Adding Batch Normalization Layer • Change learning rate

• Using gradient clip • different loss functions • Change architecture

CS B

Andrew NG Tips

Reducing (avoidable) bias & variance:



Convolutional Neural Network

Motivation: Train fully connected network is too expensive.

↳ Shallower layer: connections between close neighbors

We want: Translational Equivariance

When the input is translated by one amount, the output feature map is translated by the same amount

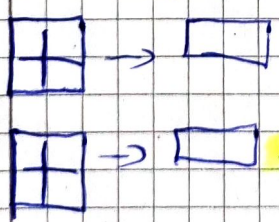
Convolutional layer:

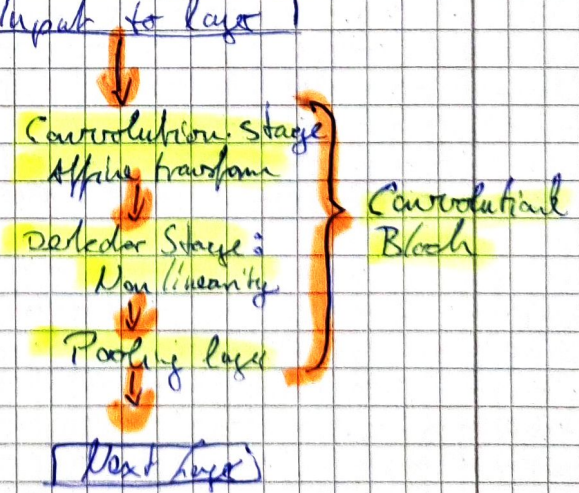
Input $n \times n$ Filter/Kernel (h, h) $k \times k$

Stride = how much you want to slide each time

Padding = how many rows/columns do you want to add to each side of the boundary

Max Pooling and Avg Pooling:





CSB

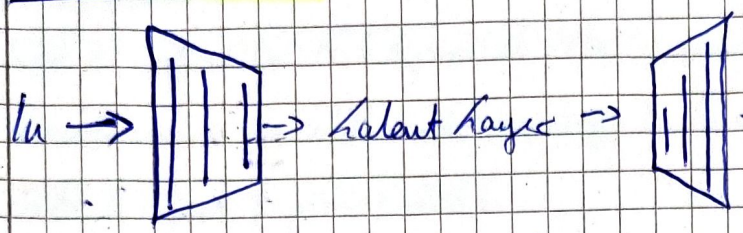
Neural Networks

VGG and Resnets

Residual connection / shortcut connection helps with the vanishing/exploding gradient problem

Autoencoders

// unsupervised



Reconstruction Loss = MSE (Input - Output)

Sometimes latent representation of data points can show interesting properties.

For denoising autoencoder, we add noise to data before feeding into encoder

↳ Assumption for Autoencoders to work:

↳ higher-dimensional data needs a representation in lower dimension

↳ data should have irrelevant dimensions (discarded by autoencoder)

CSB

Neural Networks

Sequence Models:

Motivation: For sequence data, eg: Speech

Recurrent NN, Attentions and Transformers

- In & Out lengths are not fixed
- Many inputs to one output
- The output of the first segment of data can affect the second one
- Transformers in addition to memory have attention, i.e. to which word pay more attention

Graph Neural Networks:

Motivation: Limit connection to connected neighbours.

Biological networks are graphs.

↳ node - level prediction ↳ Graph-level prediction

↳ Edge - level prediction Laplacian = degree matrix - adjacency matrix

GANS (Generative Adversarial Networks)

Motivation: Generate something from noise or a given class level or enhance the resolution of already existing ones

eg: generating synthetic genomic data

Two NN: Generator, Discriminator

- The generator takes Gaussian noise as input and generates a synthetic output of the same dimension as for the real one.
- Both synthetic and real data are passed to the discriminator. He tries to distinguish in between them
- Each epoch generator tries to generate more realistic samples to fool the generator but discriminator also gets better at discriminating real vs fake

CSB

Why is GANS harder to train?

→ Training instability

→ Mode collapse: generator fails to capture full diversity of the real data

→ difficult evaluation

CSB

1) Deep Learning

a) Backpropagation:

→ performs a backward pass to adjust the model's parameter, aiming to minimize the mean squared error

→ predicted results are compared to the correct result and error is computed → used to update the neuron parameters. Updates are propagated backwards through the network.

b) Dropout:

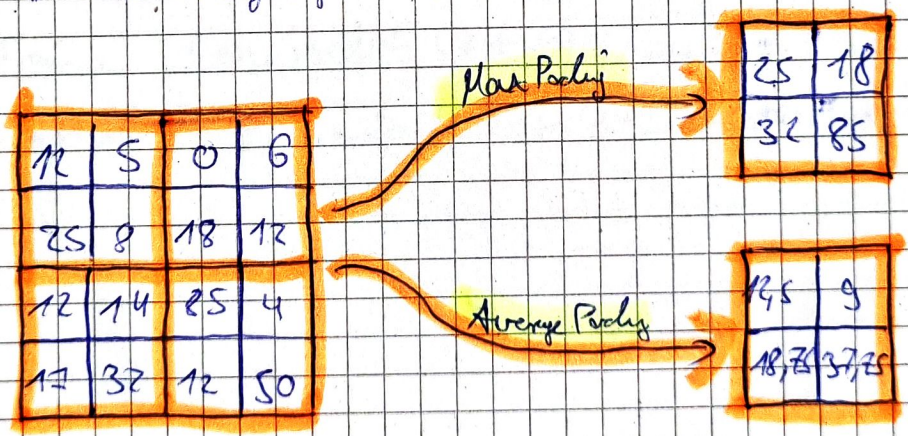
→ a regularization technique

→ ~~Deactivates~~ randomly deactivating (=0) a percentage of neurons in a NN during training

c) Pooling Layer

→ reduce the spatial dimensions of the input volume, while retaining important features

d)



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(2) Deep learning

What is the model doing at the epochs?

Forward pass → Loss calculation → Backward pass → Parameter Update

↑ (loop back)

learning faster in earlier epochs.

Python:

Concat Data Sets: data = pd.concat([d1, d2], axis=0)

Train Test Data: train_test_split(X, y, random_state=42, stratify=y)

Sequential NN:

cnn = Sequential()

cnn.add(layers X)

eg: Conv1D, Dropout, MaxPooling1D, Flatten, Dense

cnn.compile()

cnn.plot()

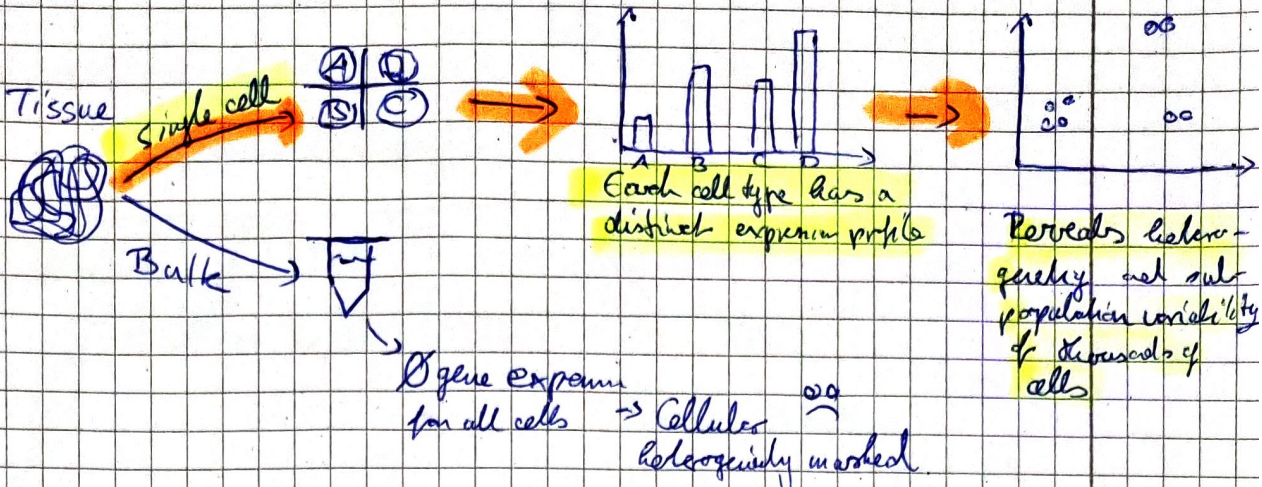
Fit Training Data: cnn.fit(X_train, ...)

Train: cnn.predict(test_x)

Confusion Matrix = confusion_matrix → accuracy_score()

CSB

Single Cell RNA Seq



→ Ø gene expression for all cells → Cellular heterogeneity masked

Single cell:
cell specific drugs!

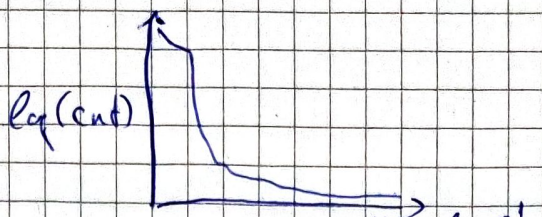
What we want for single cell:

- → universell (eg cell size)
- → in situ measurements
- → No minimum input # of cells
- → Every cell is assayed
- → Every transcript in every cell is detected
- → transcript is identified by full-length sequence
- → No doublets
- → Multimodal
- → Cost effective
- → Easy to use
- → Open Source

=> there are many single cell RNA-seq protocolls.

CS3

Droplet - based Method



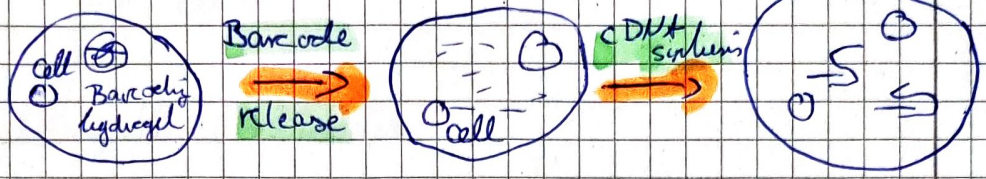
→ are a 3' technology

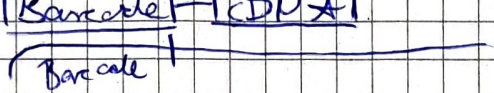
↳ more information there ↗

In Drops:

→ → → → → → → → → → →

1) → One Drop contains: One Cell,
One Barcode

2) 

3) 

Droplet Tuning Concepts

→ load beads into drops at Poisson rate μ

→ load cells into drops at Poisson rate λ

$P(\text{droplet has } j \text{ beads}) = e^{-\mu} \frac{\mu^j}{j!}$

$P(\text{droplet has } k \text{ cells}) = e^{-\lambda} \frac{\lambda^k}{k!}$

→ Cell Capture rate: the probability that a drop has at least one ~~cell~~ bead $= 1 - e^{-\mu}$

→ Cell Duplication Rate: the rate at which captured single cells are associated with two or more barcodes

$$= \frac{(1 - e^{-\mu} - \mu e^{-\mu})}{1 - e^{-\mu}}$$

↳ tradeoff between the two rates!

1 Barcode = 1 cell (optimal)

CSB

Barcode Collisions

-> Barcode Collisions : arise when two cells are separately encapsulated with beads with same barcode

-> Barcode Collision Rate : the expected proportion of assayed cells that did not receive a unique barcode

N assayed cells, M barcodes

$$1 - \frac{E(\text{cells with unique barcode})}{\#\text{ cells}} = 1 - \left(\frac{1}{e}\right)^{N/M}$$

↳ leads to synthetic doublets

↳ avoidable by small ratio $\frac{N}{M}$, relative barcode diversity high

↳ // 1000 times larger

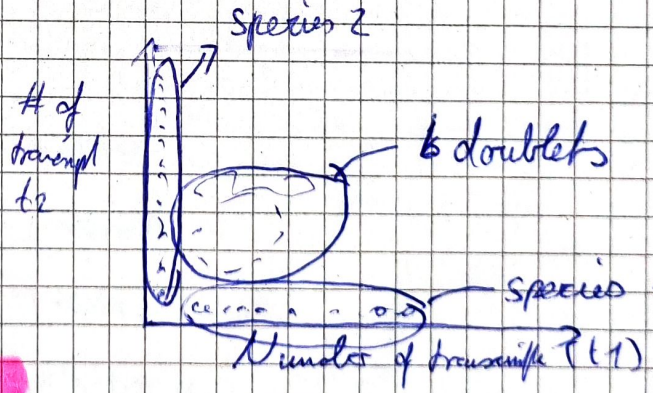
Technical Doublets

-> Technical Doublet : two or more cells inside one drop with a bead

Rate : $\frac{1 - e^{-\lambda} - \lambda e^{-\lambda}}{1 - e - \lambda}$ // fewer fast

Doublet Detection

Barnyard plot :



Biological Doublets

Biological Doublets arise when two cells form a discrete unit that does not breach apart during dissociation to form a suspension

↳ not detected via barnyard plot

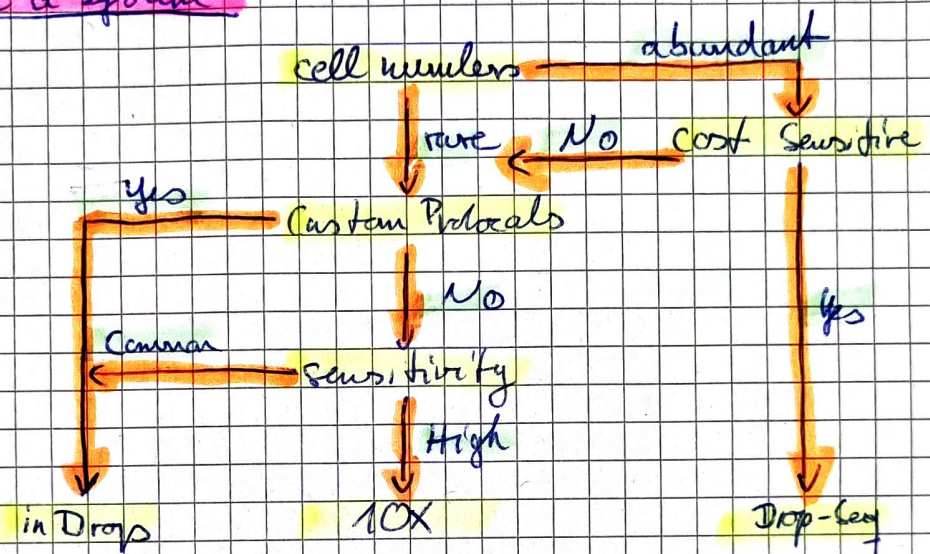
CSB

Unique Molecular Identifiers (UMI)

UMI collapsing : refers to the process of using UMIs to avoid double counting molecules after sequencing

↳ naive UMI collapsing :
just counts reads that have the same UMI, cell barcode and whose cDNAs align to the same location, as a single event
(more sophisticated UMI collapsing allows diff. locations)

Choose a system



Preprocessing layers

1) Measured → raw, Normalized
2) Corrected → technical correction, biological correction
3) Reduced → Feature Selected, Dimensional reduced

Quality Control

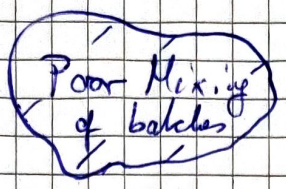
The distributions of these QC covariates are examined for outlier peaks that are filtered out.
These outlier barcodes can correspond to dying cells or doublets.

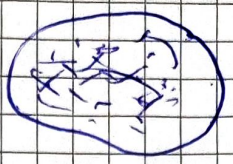
→ Barcodes with a low count depth, few detected genes, and a high fraction of mitochondrial counts are indicative of cells whose cytoplasmic mRNA has leaked out through a broken membrane and thus only mRNA located in the mitochondria is still conserved

→ Cells with unexpectedly high counts and a large number of detected genes may represent doublets
↳ high count depth thresholds are commonly used to filter out potential doublets

CSB

Batch Effect Correction





no batch correction

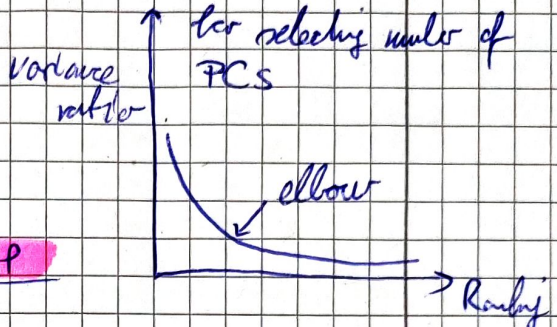
batch correction

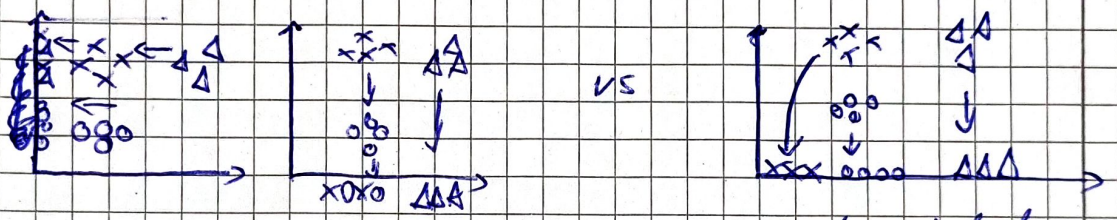
eg: ComBat for batch correction

Visualization Methods

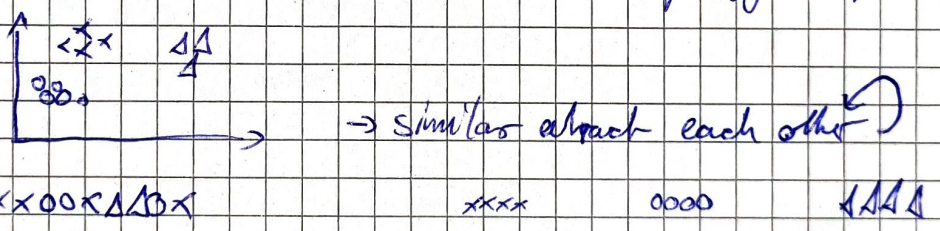
PC, tSNE, DC, UMAP, FA

Dimensional Reduction using t-SNE or UMAP





Idea:



CSS

t-Distributed Stochastic Neighbor Embedding
t-SNE

→ technique for dimensionality reduction
→ unsupervised and non-linear
→ preserving local similarities (small distances)
→ Only Visualization, No Clustering!

1) Step: select neighbours w.r.t Gaussian Distribution over points in high dimensional space
2) Step: select neighbours w.r.t t-distribution over points in low dimensional space
3) Minimize Kullback-Leibler divergence between both distribution

---

1) Probability to choose x_i as neighbor of x_j p_{ij}
2) Heavier tails allow better modeling of larger distances.
   Probability to choose y_i as neighbor of y_j q_{ij}
3) use gradient descent $\frac{\delta C}{\delta y_i}$, $C = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$

CSB

Uniform Manifold Approximation and Projection (UMAP)

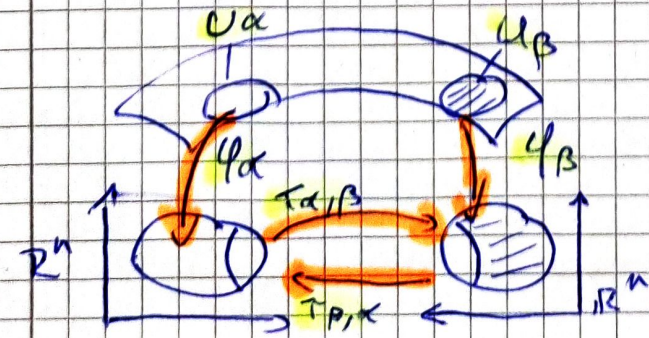
- technique for dimensional reduction
- unsupervised and non-linear
- Preserving local similarities (small distances)
- Preserving global dissimilarities (large distances)
- Scales better than t-SNE computationally
- Only visualization, No clustering!

0-simplex •
1-simplex
2-simplex
3-simplex

1) approximate manifold for data in high dimensional space using simplical complexes as neighborhood graph
2) approximate distances for data in low dimensional space using spectral embedding
3) Optimize low dimensional fuzzy topology to be similar to high dimensional fuzzy topology via fuzzy set cross entropy

---

1) A k-simplex is a convex hull of k+1 points, can be used as building blocks for topological spaces and covers
- constructing an open cover as simplical complex
  - Consider vicinities around points and determine overlaps
    - represent overlaps by simplices to get simplical complex
      - disconnected, since data are not uniformly distributed
- assume data is uniformly distributed on the manifold, but notion of distances changes along the manifold => Riemannian Metric



CSS

UMAP

- extend radius to k-nearest neighbors
  - small k focuses on local topology
  - greater k focuses on global topology
- fuzzy topology allows local distance measures but may still suffer from isolated points in high dimensions
- In high dimensions distance to first neighbor is large but similar as to others
- Focus on the differences in distances among nearest neighbors rather than the absolute distance.
- enforce local connectivity by getting fuzzy only after first neighbor

CSS

① Single-cell Technology

The cell capture rate is the probability that
a droplet has at least one ~~cell~~ bead ~~droplet~~

The cell duplicability rate is the rate
at which captured single cells are associated
with two or more different barcodes

UMIS
are used to identify individual molecules

② Barcodes

a) Main Job of a barcode in Single Cell RNA-Seq?
Barcodes: play a crucial role in identifying and distinguishing
individual cells and their corresponding RNA molecules.

③ Dimensional Reduction

First steps of t-SNE is to
select neighbors w.r.t. Gaussian Distribution over
points in high-dimensional space

The first step of of UMAP ~~and~~ is to
approximate a manifold in high-dimensional data using simplest Complexes

t-SNE and UMAP are used
for visualization

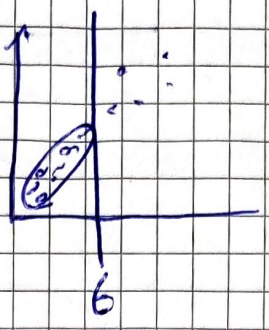
non-linear methods: t-SNE, UMAP

both local and global dissimilarities: UMAP

④ Scanpy

→ Filter out outliers → doublets or
→ dying cells

→ Set threshold such that outliers are cut out



→ Why do we only perform the downstream analysis with highly variable genes?

Downstream Analysis

Cell level { Cluster Analysis: Clustering, Composition analysis, Cluster annotation
Trajectory Analysis: Trajectory inference, Metastable states, Gene expression Dynamics

Gene level: Gene Set Analysis / Gene Regulatory Networks / Differential expression analysis

Steps:
1) Data preprocessing
2) Quality Control
3) Normalization
4) Dimensionality Reduction
5) Clustering
6) Differential expression analysis
7) Visualization & Interpretation

Used for:
aims to unravel heterogeneity, identify novel cell types, characterize cell states.
↳ identifying potential therapeutic targets

→ Biological relevance, Noise Reduction, Computational Efficiency, Interpretability, Resource optimization

Molecular Evolution & Phylogeny

Evolution is a process of mutation with selection

Molecular evolution is the study of changes in gene and protein throughout different branches of the tree of life

Phylogeny is the inference of evolutionary relationships.

Cladogram

→ Answers: Is my favorite gene under selective pressure?

Phylogeny: the doctrine of phylogenetic evolution of living beings, the emergence of species and their evolutionary relationships

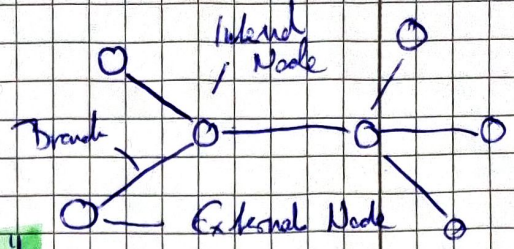
Anagenesis: Transformation of species

Cladogenesis: Species splitting

Taxon: Groups of living beings, Unity with biological classification

CS B

Phylogenetic Trees



→ Topology and branch lengths

→ Internal Node = Hypothetical Taxonomic Unit

↳ common ancestors ↳ Based on predictions

→ External Node = OTU

↳ Now living species ↳ Sequenced

→ Branch = Evolutionary Kinship

⇒ Can also be rooted

Root: ∠ LUCA

→ Acyclic Structure

→ Internal nodes in a fully resolved phylogenetic tree: 2 offspring / 1 ancestor

→ Evolutionary relationship

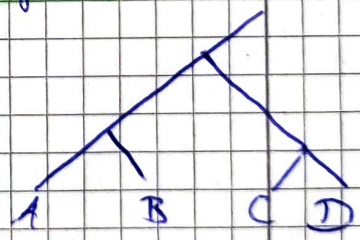
→ LUCA unknown

→ No clear evolutionary direction

} unrooted

↳ rooted: clear time course of evolution or divergence

Cladogram

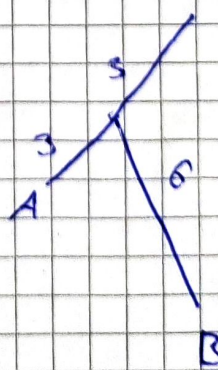


→ rooted / unrooted

→ Only ancestry is displayed

→ Edge length has no meaning!

Additive tree



→ rooted / unrooted

→ Edge: Length = quantitative measure of evolution

Sum of edge length = evolutionary distance of two taxa

No information about mutation rate.

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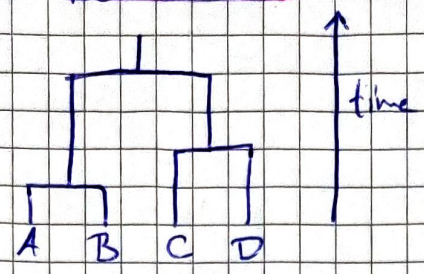
Ultrametric Tree



→ rooted

→ edges constant mutation rate

→ Identical ~~[illegible]~~ evolutionary distance between ancestor and their descendants

Scaling:

→ Evolutionary Time = Real Time Intervals

→ Evolutionary Distance = Measure of Distance between Taxa

Complexity: Phylogenetic Trees

n leaves

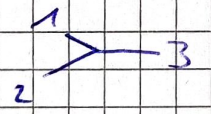
	unrooted	rooted
# Internal nodes	$n-1$	$n-2$
# nodes	$k = 2n-2$	$k = 2n-1$

Completely resolved phylogenetic tree

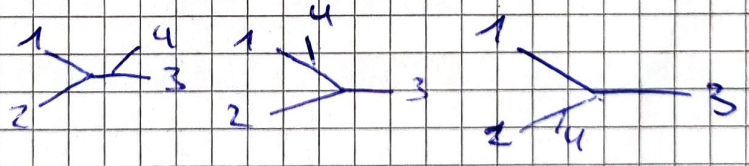
$\Leftrightarrow$ binary tree with n leaves

Root can be attached to any branch $\Rightarrow$ $2n-3$ rooted trees for unrooted tree

$n = 3$ leaves:



$n = 4$ leaves:



possible trees:

unrooted: $\frac{(2n-5)!}{(2^{n-3}(n-3)!)}$

rooted: $\frac{(2n-3)!}{(2^{n-2}(n-2)!)}$

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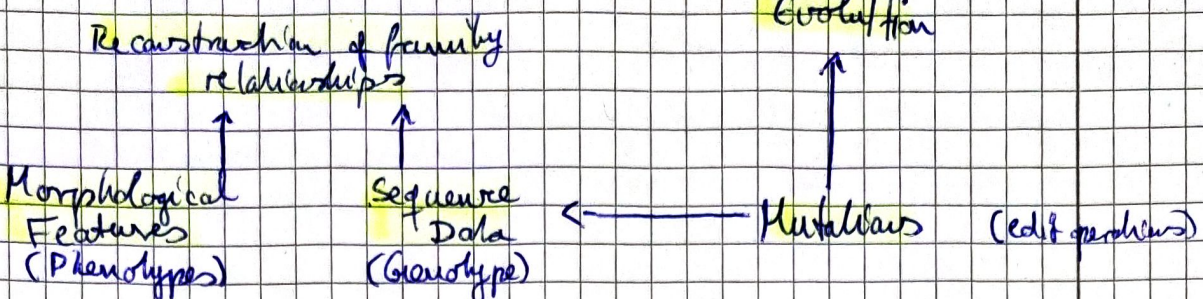
Molecular Evolution

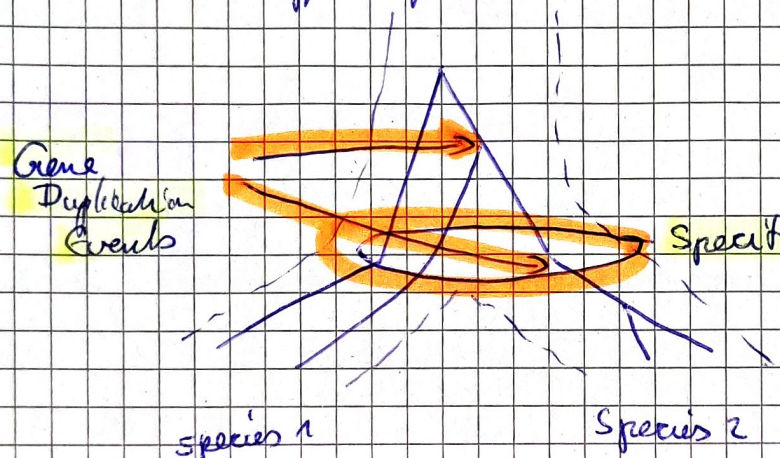
A measure of evolutionary distance:



Gene Tree / Species Tree

Gene Trees focus on the evolution of a specific gene within populations.

Species Tree depict the evolutionary relationships between different species.



A gene may duplicate before or after two species diverge!

Main cause of evolutionary change is random drift of mutant alleles that are selectively neutral

For every given protein, the rate of molecular evolution is approximately constant in all evolutionary lineages.

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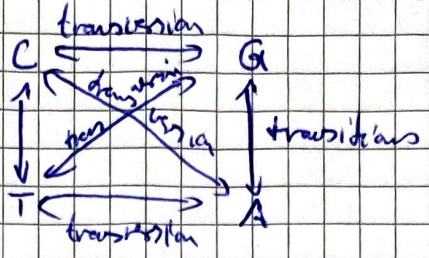
Acceptance of Mutations

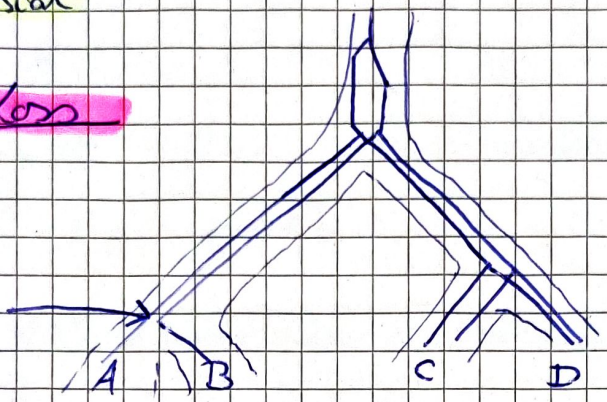
Success of a mutation depending on its effect on organism.

2 types of substitution



- Transitions are more common.
- Transversion

Gene Loss



Horizontal Gene Transfer: Gene transfer between species

Modelling of Molecular Evolution

Starting Point: Sequence Differences = Evolutionary Changes

Looking For: A measure of the evolutionary distance between two sequences

Ideally: # of mutations since last ancestor (not possible)

Solution: Estimation Model, based a correct alignment of homologous sequences

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P-Distance

Share of non-identical aligned positions $p = \frac{D}{L}$

D : # not identically aligned positions

L : # aligned positions (without gaps)

↯ : underestimates the actual # mutations

Poisson Distance

Assume: # Mutations is Poisson Distributed

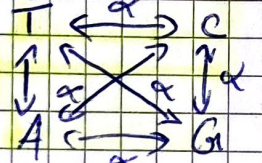
$d_p = -\ln(1-p)$, p: P-Distance

Distribution of the number of events that occur at a constant rate and independent of each other in a fixed time interval

Jukes Cantor 1-Parameter Model

Substitution Rate Matrix.

All positions are independent and have identical mutation rate α $\forall$ possible transitions



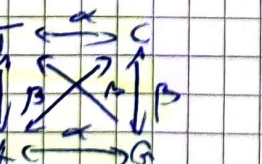
$$Q^{JC} = \begin{array}{c} \\ A \\ C \\ G \\ T \end{array} \begin{array}{c} \begin{array}{cccc} A & C & G & T \end{array} \\ \begin{pmatrix} -3\alpha & \alpha & \alpha & \alpha \\ \alpha & -3\alpha & \alpha & \alpha \\ \alpha & \alpha & -3\alpha & \alpha \\ \alpha & \alpha & \alpha & -3\alpha \end{pmatrix} \end{array}$$

Row sum = 0 $\Leftrightarrow$ Sequence length constant

Column Sum = 0 $\Leftrightarrow$ Base composition constant

Kimura 2-Parameter model

→ Different mutation rates for transitions α and transversion β



$$Q^{K2P} = \begin{array}{c} \\ A \\ C \\ G \\ T \end{array} \begin{array}{c} \begin{array}{cccc} A & C & G & T \end{array} \\ \begin{pmatrix} -2\beta-\alpha & \beta & \alpha & \beta \\ \beta & -2\beta-\alpha & \beta & \alpha \\ \alpha & \beta & -2\beta-\alpha & \beta \\ \beta & \alpha & \beta & -2\beta-\alpha \end{pmatrix} \end{array}$$

$$d_{K2P} = -\frac{1}{2}\ln(1-2P-Q) - \frac{1}{4}\ln(1-2Q)$$

P = Share of sequence positions that have a transition

Q = Share of sequence positions that have a transversion

$p = P + Q$

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Assumptions of Jukes - Cantor & Kimura

- → only substitutions
- → position independent
- → probabilities only depend on the current state
- → same mutation rate $\forall$ positions
- → 1:1:1:1 base composition

@ Model is not always apply to real data!

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Reconstruction of phylogenetic tree

1) Data Selection
2) Model Molecular Evolution
3) Method of tree reconstruction
4) Evaluation

Data Selection

Ideally: → Sequence occurs in all investigated species in single copy
→ No horizontal gene transfer
→ Mutation Rate: High separating closely related species
Low separating far-away species
→ Sequence with highly conserved and very variable sections

Start: a correct alignment, sequence should align at full length

Methods

Distance-based / Clustering

→ MSA determines distance between 2 pairs of OTU
→ Branched together based on distances
→ exactly one tree produced
→ very large data
→ multiple trees for one alignment
→ Comparison of diff. trees difficult
→ Minimal Evolution
→ Max Likelihood

Alignment-based / Topology

→ Multiple Alignment used directly
→ Finding the optimal topology by evaluating Alignment positions
→ All trees that can explain the data are generated
→ Expensive
→ Can be compare trees
→ Max Parsimony

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UPGMA

Unweighted Pair Group Method using arithmetic mean:

→ distance based

→ Principle: sequential clustering

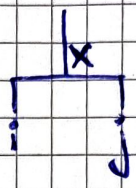
→ Assume: Molecular clock, same AA substitution in ∀ seq since LUCA

→ Result: rooted ultrametric tree

→ 1) New internal cluster

Merge ~~Find~~ nearest related sequences/clusters i and j ($\min(d_{ij})$) into a cluster X.

Cluster with a height $\frac{d_{ij}}{2}$ →



2) Update Distance Matrix

Replace the seq's/clusters i and j with X in D and recalculate the distances.

Distance of two clusters X and Y: $d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$

→ Trees with molecular clock (MCP)

⇔ all leaves end at the same position, each path root → leaf = same time

⇔ Trees with MCP are ultrametric

⇔ triangle inequality

⇔ any three leaves, i, j, k can be named such:

$d_{ij} \leq d_{jk} = d_{ki}$

three point condition

Neighbor-Joining

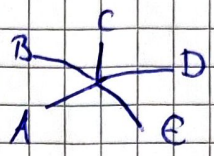
distance-based, considers different mutation rates

Principle: Sequentially group OTUs while minimizing branch lengths

Assumption: Minimum evolution, i.e. tree with the minimal sum of branch lengths is the best

↳ NJ provides an approximation

Result: unrooted additive tree



0) Initialization: Star-Shaped tree

1) Find nearby OTUs

a) compute M, $M_{ij} = (n-2)\, d_{ij} - \sum_{k \neq i} d_{ik} - \sum_{k \neq j} d_{jk}$

$\sum_{k \neq i} d_{ik}$: ↳ Total differences between i and all others

b) choose smallest M_{ij} and define a newer node Y with branch lengths

$$b_{iY} = \frac{1}{2} d_{ij} + \frac{1}{2(n-2)} \left(\sum_{k \neq i} d_{ik} - \sum_{k \neq j} d_{jk} \right)$$

$$b_{jY} = d_{ij} - b_{iY}$$

2) Update distance matrix

$$d_{YK} = \frac{1}{2} (d_{iK} + d_{jK} - d_{ij})$$

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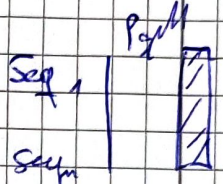
Maximum Parsimony

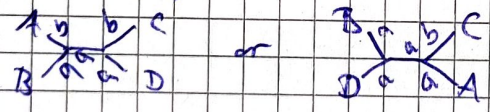
Alignment-based Method.

Tree that requires the least number of evo. changes to explain the observed differences

→ ∀ topologies are considered
→ results in multiple unrooted trees
→ no explicit evolution model
→ based on minimal evolution
→ no distance matrix, but counts state changes
→ Not all positions help distinguish topologies
→ Position informative, if
  → ≥ 2 different nucleotides
  → occur ≥ 2 times



Alg.:
→ search all informative positions
→ Determine which tree topologies need to be examined
→ For each position determine the lowest number of substitutions



⇒ Tree that requires the least number of substitutions is the maximum parsimony tree

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Locate the root

In : Unrooted Tree

→ provide outgroup = faraway-group

→ Root between outgroup and remaining taxa

Bootstrap Tree Assessment

→ estimate significance of the branches of a tree

Alg

1- Modify original data : • new dataset by column permutations of the alignment.

• omit columns

- Reconstruct Tree

→ How often do modified data produce the same results as the original dataset = bootstrap value

for each internal branch

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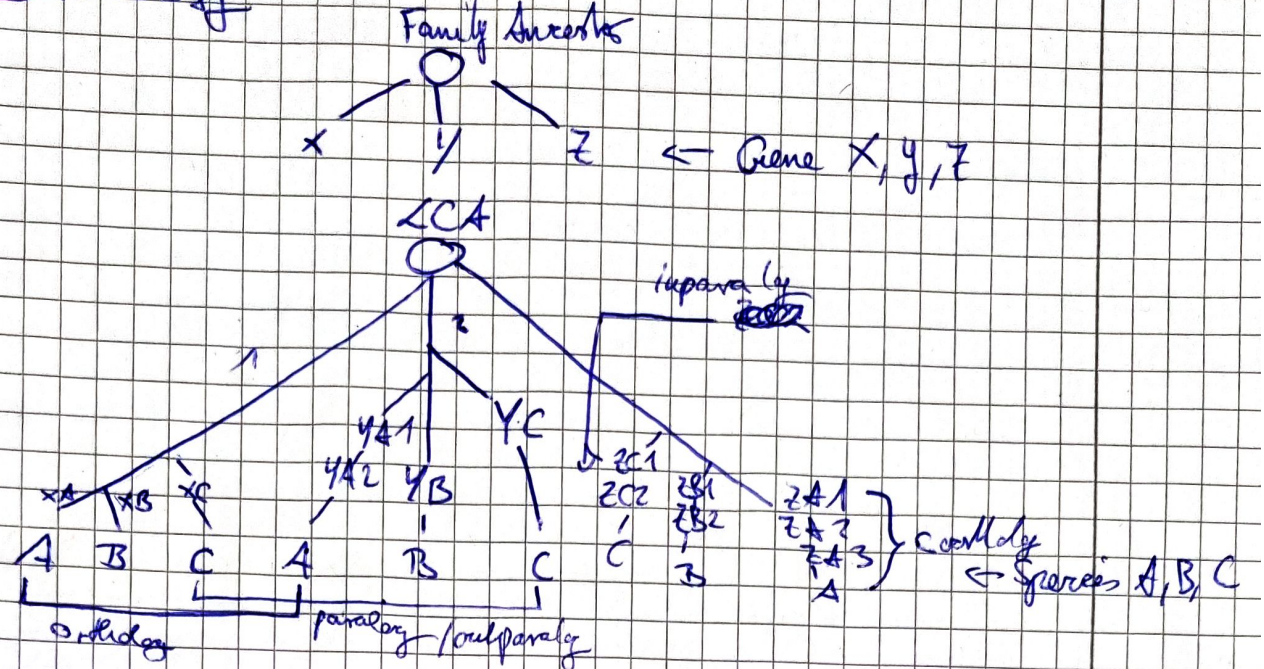
① Homology



Orthology: are genes in different species that evolved from a common ancestral gene through speciation. ✓

related by Speciation Event

eg: XA, XB, XC — same gene, diff species

Paralog: are genes that arise from gene duplication events within a single species. ✓

related by gene duplication event

eg: XB, YB, ZB — same species, diff gene

Inparalog: are genes that arise from gene duplication events within the same genome. ✓

duplication after Speciation Event

eg: ZA1, ZA2, ZA3

Outparalog: are paralogous genes that arose from gene duplication events in different species.

duplication before Speciation Event

Share common ancestral gene that are or more copies were lost in one of the lineage

eg: XC, YC, ZC

Coorthologs: are orthologous genes that are retained in multiple copies within the same genome. ✓

genes related by Speciation that underwent subsequent duplication

eg: (ZA1, ZA2, ZA3) & (ZB1, ZB2) ✓

(1)

b) Early evolution of genes:

→ Mutation

→ Genetic recombination

→ Gene Flow

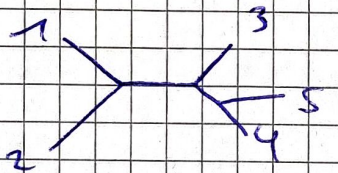
(2)

a) 5 Taxa

unrooted trees if $n=5$: $\frac{(2n-5)!}{2^{n-3}(n-3)!} = 15$

rooted trees $n=5$: $\frac{(2n-3)!}{2^{n-2}(n-2)!} = 105$

unrooted tree with 4 taxa ⇒ 3 taxa



(and all permutations of 1,2,3,4)

b)

Gene tree focuses on the evolution of specific genes within populations

Species Tree depict the evolutionary relationships between different species

⇒ Not necessary the same

(3) Quantitative Models of molecular evolution

a) Evolutionary Model

Models in lecture:

- P-Distance //
- Poisson Distance // constant rate, independent, fixed time interval
- Jukes-Cantor // all positions are independent with identical mutation rate α
- Kimura // different mutation rate for transition/transversion

→ only replacements
replacement is position independent
rate only depends on the current state
same rate $\forall$ positions
all cases

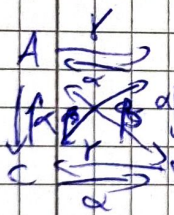
Not presented: Felsenstein, K2P

b) Normed

$$Q^{JC} = \begin{array}{c} \\ A \\ C \\ G \\ T \end{array} \begin{array}{c} \begin{array}{cccc} A & C & G & T \end{array} \\ \begin{pmatrix} -3\alpha & \alpha & \alpha & \alpha \\ \alpha & -3\alpha & \alpha & \alpha \\ \alpha & \alpha & -3\alpha & \alpha \\ \alpha & \alpha & \alpha & -3\alpha \end{pmatrix} \end{array}$$

→ Now we want to consider a possible bias in the GC content

Substitution Matrix:



$$Q^{sub} = \begin{array}{c} \\ A \\ C \\ G \\ T \end{array} \begin{array}{c} \begin{array}{cccc} A & C & G & T \end{array} \\ \begin{pmatrix} \cdot & \gamma/2 & \gamma/2 & \alpha/2 \\ \cdot & \cdot & \alpha/2 & \gamma/2 \\ \cdot & \alpha/2 & \cdot & \gamma/2 \\ \alpha/2 & \gamma/2 & \gamma/2 & \cdot \end{pmatrix} \end{array} \begin{array}{l} = 0 \\ = 0 \\ = 0 \\ = 0 \end{array}$$

$\gamma = \pi_G + \pi_C$

c) with different for transversion and transition

$$Q = \begin{array}{c} \\ A \\ C \\ G \\ T \end{array} \begin{array}{c} \begin{array}{cccc} A & C & G & T \end{array} \\ \begin{pmatrix} -2\gamma-\beta & \gamma & \gamma & \beta \\ \beta & -2\beta-\beta-\alpha & \beta & \alpha \\ \alpha & \beta & -2\beta-\beta-\alpha & \beta \\ \beta & \gamma & \gamma & -2\gamma-\beta \end{pmatrix} \end{array} \begin{array}{l} = 0 \\ = 0 \\ = 0 \\ = 0 \end{array}$$

$\gamma > \alpha$

$\gamma > \beta$

$$\begin{array}{c} \\ A \\ C \\ G \\ T \end{array} \begin{array}{c} \begin{array}{cccc} A & C & G & T \end{array} \\ \begin{pmatrix} * & \kappa\gamma/2 & \gamma/2 & (1-\gamma)/2 \\ \kappa(1-\gamma)/2 & * & \gamma/2 & (1-\gamma)/2 \\ (1-\gamma)/2 & \gamma/2 & * & \kappa(1-\gamma)/2 \\ (1-\gamma)/2 & \gamma/2 & \kappa\gamma/2 & * \end{pmatrix} \end{array}$$

(3)

c) Use MSA to create a distance matrix D

$$P = \frac{D}{L}$$ // aligned without gaps

$$D = \begin{array}{c} \\ S1 \\ S2 \\ S3 \\ S4 \end{array} \begin{array}{c} \begin{array}{cccc} S1 & S2 & S3 & S4 \end{array} \\ \begin{pmatrix} p(S1,S1) & p(S1,S2) & p(S1,S3) & p(S1,S4) \\ p(S2,S1) & p(S2,S2) & p(S1,S2) & p(S2,S4) \\ p(S3,S1) & p(S3,S2) & p(S1,S3) & p(S3,S4) \\ p(S4,S1) & p(S4,S2) & p(S1,S4) & p(S4,S4) \end{pmatrix} \end{array}$$

eg.

$$p(S1,S2) = \frac{9}{50}$$

CSB

⑨ UPGMA

1) min d_{ij}
   altezza $\frac{d_{ij}}{2}$
2) Update $d_{xy} = \frac{1}{N_x N_y} \sum_{i \in x, j \in y} d_{ij}$

	A	B	C	D	E
B	2				
C	4	4			
D	6	6	6		
E	6	6	6	4	
F	8	8	8	8	8

$A \cup B = U$

$d_{CU} = \frac{1}{2}(4+4) = 4$
$d_{DU} = 1/2\,(6+6) = 6$
$d_{EU} = 1/2\,(6+6) = 6$
$d_{FU} = 1/2\,(8+8) = 8$

	U	C	D	E
C	4			
D	6	6		
E	6	6	(4)	
F	8	8	8	8

$D \cup E = W$

$d_{WC} = 1/2\,(6+6) = 6$
$d_{WU} = \frac{1}{4}(6+6+6+6) = 6$
$d_{FW} = 1/2\,(8+8) = 8$

	V	C	W
C	(4)		
W	6	6	
F	8	8	8

$U \cup C = X$

$d_{XW} = \frac{1}{6}(6+6+6+6+6+6) = 6$
$d_{FX} = \frac{1}{3}(8+8+8) = 8$

	X	W
W		
F	8	8